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FULL STACK APPLICATION DEVELOPMENT

#SKILL - 4: Spring Dependency Injection – Constructor & Setter Injection

Prerequisites:

• Basic knowledge of Java

• General Idea on Spring Framework

• Basic Idea on DI & IoC

Tasks:

1. Create Student class with fields: studentId, name, course, year.

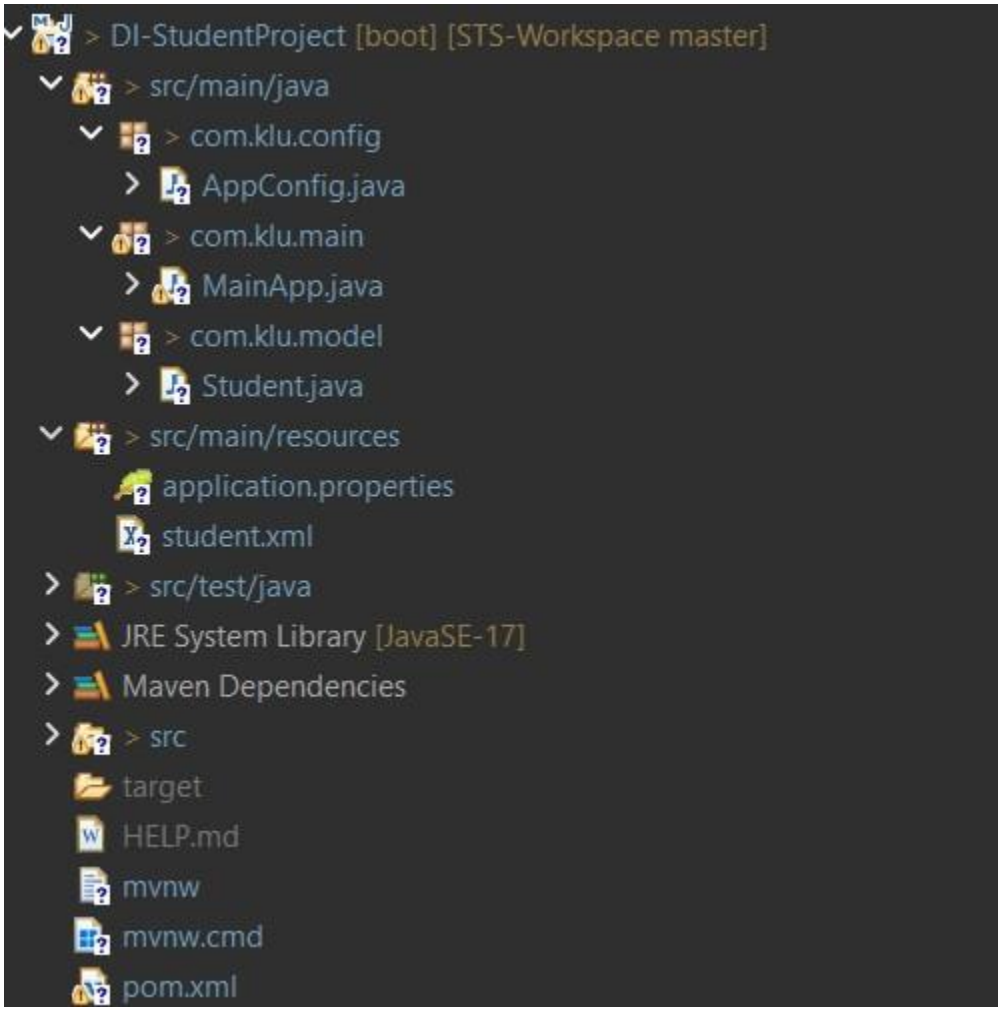
2. Implement constructor injection.

3. Implement setter injection.

4. Configure using XML and Annotations.

5. Load Spring IoC container and retrieve bean.

Structure:



AppConfig.java:

```
package com.klu.config;


import org.springframework.context.annotation.ComponentScan;

import org.springframework.context.annotation.Configuration;

import org.springframework.context.annotation.ImportResource;
```

```
@Configuration
@ComponentScan(basePackages = "com.klu.model")
@ImportResource("classpath:student.xml")   // IMPORT XML INTO JAVA CONFIG
public class AppConfig {
}
```

MainApp.java:

```
package com.klu.main;

import org.springframework.context.ApplicationContext;
import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.klu.config.AppConfig;
import com.klu.model.Student;

public class MainApp {

   public static void main(String[] args) {

      ApplicationContext context =
            new AnnotationConfigApplicationContext(AppConfig.class);

      // Annotation-based bean
      Student s1 = (Student) context.getBean("annoStudent");
      s1.display();
```

```
        System.out.println(" ------------------");

        // XML-based bean
        Student s2 = (Student) context.getBean("xmlStudent");
        s2.display();
    }
}
```

Stident.java:

```
package com.klu.model;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;

@Component("annoStudent")
public class Student {

    private int studentId;
    private String name;
    private String course;
    private int year;

    // Constructor Injection
    public Student(@Value("101") int studentId,
                   @Value("Lasyadeep") String name,
                   @Value("B.Tech") String course,
                   @Value("2") int year) {
```

```
        this.studentId = studentId;
        this.name = name;
        this.course = course;
        this.year = year;
    }

    // Setter Injection
    @Value("FSAD")
    public void setCourse(String course) {
        this.course = course;
    }

    @Value("2")
    public void setYear(int year) {
        this.year = year;
    }

    public void display() {
        System.out.println("Student Details:");
        System.out.println("Student ID : " + studentId);
        System.out.println("Name       : " + name);
        System.out.println("Course     : " + course);
        System.out.println("Year       : " + year);
    }
}
```

Student.xml:

```
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    http://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="xmlStudent" class="com.klu.model.Student">
    <constructor-arg value="102"/>
    <constructor-arg value="Rahul"/>
    <constructor-arg value="Java"/>
    <constructor-arg value="1"/>

    <property name="course" value="FSAD"/>
    <property name="year" value="2"/>
  </bean>

</beans>
```

OUTPUT:

```
------------------
Student Details:
Student ID : 102
Name       : Rahul
Course     : FSAD
Year       : 2
```

VIVA QUESTIONS:

1. **What is HQL and how does it differ from SQL?**
   DI is a technique where Spring provides the required objects to a class instead of the class creating them itself.
2. **Why do we use sorting in HQL?**
   Constructor DI injects dependencies through the constructor, while Setter DI injects them using setter methods.
3. **What is pagination and where is it used?**
   The IoC container creates, manages, and injects objects (beans) in a Spring application.
4. **Name any three aggregate functions supported in HQL.**
   DI improves loose coupling, easier testing, and better code maintainability.
5. **What is the difference between HQL and HCQL (Hibernate Criteria Query Language)?**
   Yes, both constructor and setter injection can be used together in the same class.

Git Hub Link : https://github.com/hemalatharachakonda/2400030211-Skill_experiments.git